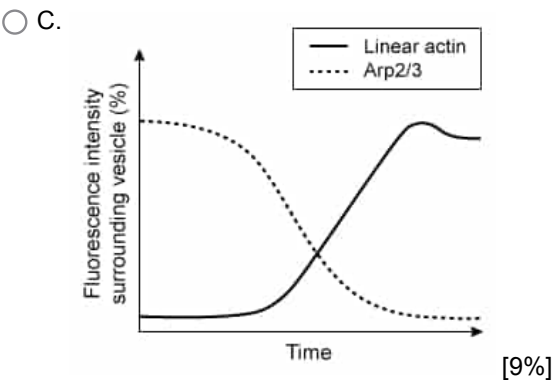
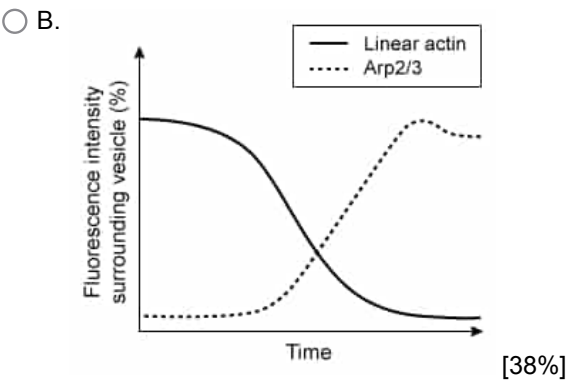
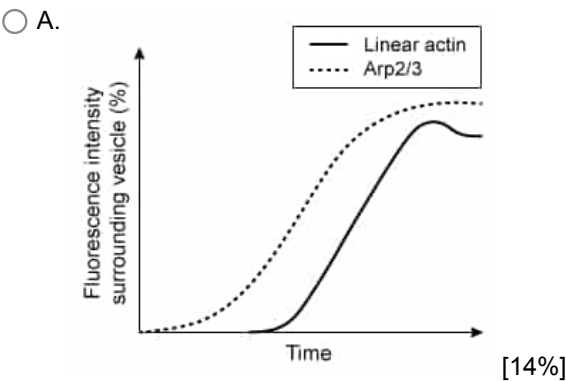
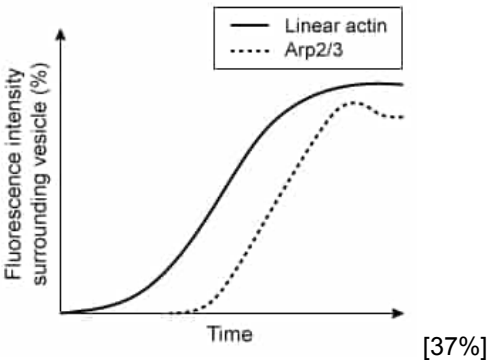


Researchers studying exocytosis in mammalian salivary gland cells hypothesize that linear actin filaments accumulate around secretory vesicles, and that the enzyme complex Arp2/3 is recruited to remodel linear actin filaments, facilitating vesicle release. In a study, researchers fluorescently label linear actin filaments and Arp2/3 in salivary gland cells undergoing exocytosis. If the hypothesis is true, which of the following graphs likely illustrates how linear actin and Arp2/3 fluorescence intensities surrounding a vesicle will change over time during exocytosis?

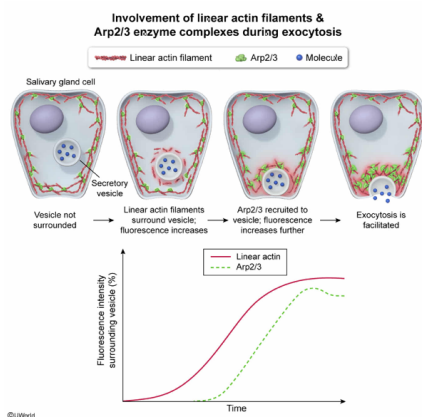


☒ D.



Correct
 38% answered correctly

Explanation:



Exocytosis, which requires energy (eg, ATP), is a form of **membrane transport** by which secretory vesicles within a cell fuse with the **plasma membrane**, allowing for the release of their contents into the extracellular environment.

In this question, researchers are studying cells of mammalian **salivary glands**, which are **exocrine glands** located in the mouth. These glands are components of the **digestive system** that secrete saliva, a liquid substance that contains **enzymes** (eg, amylase, lipase) involved in chemical digestion of food. Many types of exocrine glands, including salivary glands, use exocytosis for the secretion of molecules (eg, digestive enzymes).

The researchers hypothesize that **linear actin filaments accumulate** around secretory vesicles during this process. The enzyme complex **Arp2/3** is thought to be **recruited** to secretory vesicles to remodel these linear actin filaments, facilitating vesicle release.

In a study, the researchers **fluorescently label** linear actin filaments and Arp2/3 in salivary gland cells undergoing exocytosis. If their hypothesis is true, linear actin will likely begin accumulating (*increasing* in fluorescence) around the vesicle *first*, serving as a **substrate** for Arp2/3. Next, Arp2/3 enzyme complexes will be recruited to the vesicle and accumulate (*increase* in fluorescence) over time as they remodel these linear actin filaments. Only **Choice D** correctly illustrates this outcome.

(Choice A) Because linear actin is the *substrate* for Arp2/3 (ie, must be present for Arp2/3 to perform its function), the Arp2/3 enzyme complexes would likely accumulate *after* (*not prior to*) the start of linear actin accumulation.

(Choice B) The question states that linear actin filaments are hypothesized to *accumulate* around secretory vesicles during exocytosis. Therefore, if this hypothesis is true, the amount (fluorescence intensity) of linear actin will likely increase, *not decrease*, over time during exocytosis.

(Choice C) It is hypothesized that Arp2/3 is *recruited* to the secretory vesicle to remodel linear actin filaments during exocytosis. Therefore, the amount of Arp2/3 (ie, Arp2/3 fluorescence) surrounding the vesicle would likely *increase*, *not decrease*, during this process if the hypothesis is true.

Educational objective:

During exocytosis, secretory vesicles within a cell fuse with the plasma membrane, allowing the contents within these vesicles to be released into the extracellular environment. Exocytosis allows the release of molecules from certain exocrine glands in the body, including the salivary glands.

Time spent: 0 sec
Subject:
Biology

QID: 403126
System:
Digestion and Excretion